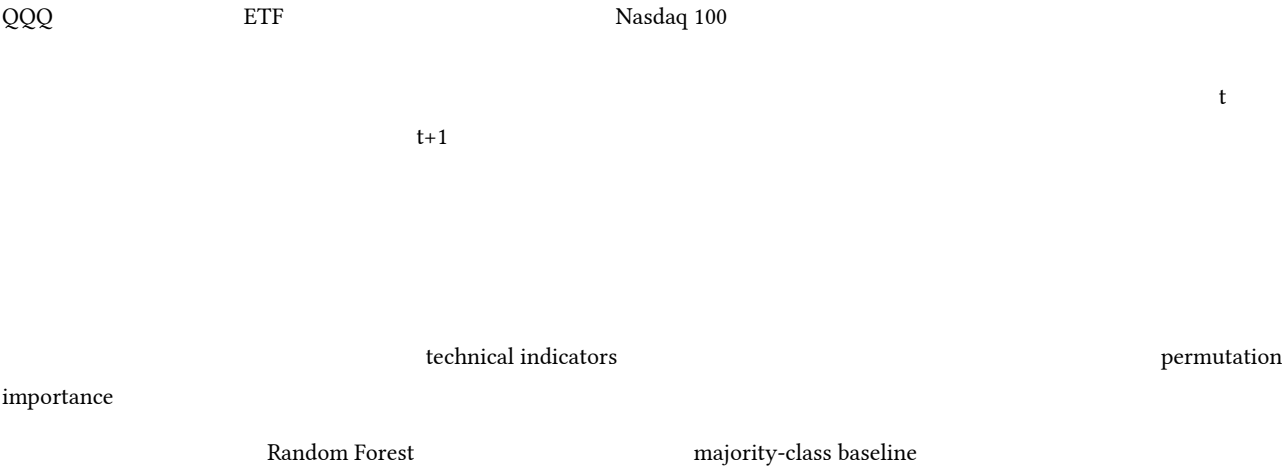


QQQ Technical Indicator Feature Importance

Big Data Analytics Project Report

Tien Hanprab		D6 900 557			
Nasdaq Historical Quotes for QQQ		4	2021	17	2026
26	2026				
					QQQ
					technical indicators 21
Nasdaq	1,433	1,412			[1]
		283	Majority baseline	accuracy 54.77%	Random Forest
51.59%	HistGradientBoosting	53.00%	baseline		
accuracy					
Permutation importance	HistGradientBoosting	atr_14_pct	(	balanced accuracy 0.0315 ±	
0.0085	)				
notebooks/pipeline_qqq_5_year.ipynb					/ pipeline 5

1



- RQ1 What are the major price, return, volume, and volatility patterns of QQQ during the configured analysis period?
- RQ2 Which technical indicators are the most important features for classifying QQQ's next-day price direction?
- RQ3 Does the Random Forest classifier perform better than a simple baseline on unseen chronological data?

target_up(t) = 1

Close(t+1) > Close(t)

0

Close(t+1) ≤ Close(t)

t+1

next_return

target

2

QQQ

Nasdaq public historical quote service

endpoint

https://api.nasdaq.com/api/quote/XXX/historical

assetclass=etf, fromdate=2021-01-01, todate=2026-09-17

limit=5000

Nasdaq XXX Historical Data

[1] raw JSON

metadata

response

	parse	
Date	datetime	
Open	float	USD
High	float	USD
Low	float	USD
Close	float	USD
Volume	integer	

Response

1,433

2021-01-04

2026-09-17

500

assignment [4] metadata

retrieval 2026-09-19T22:15:36Z, URL

SHA-256

5af8f404 421 142e426ca6 857 137b9355c4bee85fd8067ae6 923 973e3dab7001a

:

Close

response

Nasdaq

adjusted

close

total return

3

Pipeline	schema	JSON	date	MM/DD/YYYY	datetime
/			OHLCV		
			missing	non-finite	
		High/Low			
Raw response	1,433	Nasdaq			
	0	missing OHLCV			
	0				
Clean OHLCV	1,433				
Indicator warm-up	20	20			
target	1	Close(t+1)			
Model-ready	1,412	21 features	next_return	target_up	

OHLCV

: 1,433 – 20 – 1 = 1,412

feature

warm-up

pipeline

audit

2021-02-02

2026-09-16

data/processed/5_year/qqq_features.csv

4

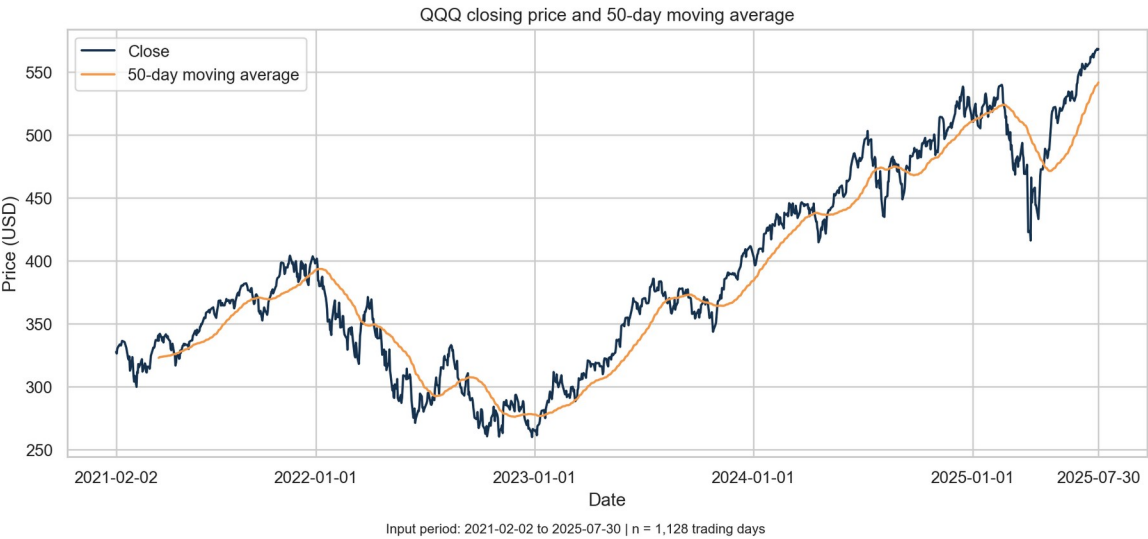
EDA

5-7

training period 2021-02-022025-07-30

1,128

holdout



1

50

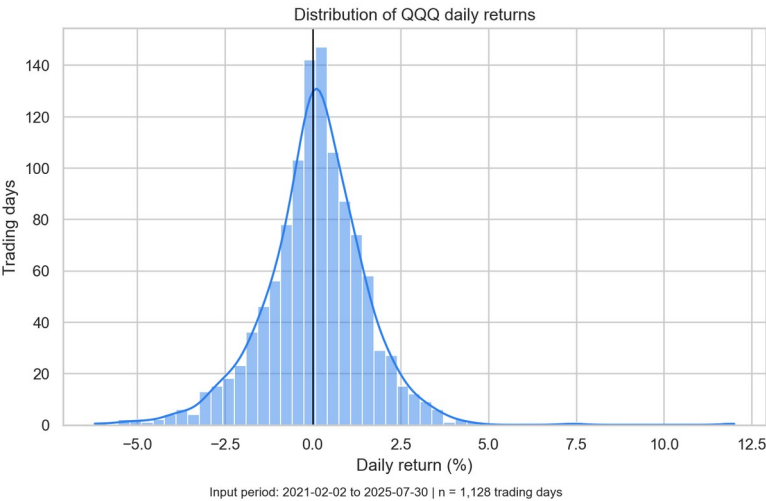
260.10

327.682022-12-28

568.02 USD568.14

(+73.35%)

2025-07-28

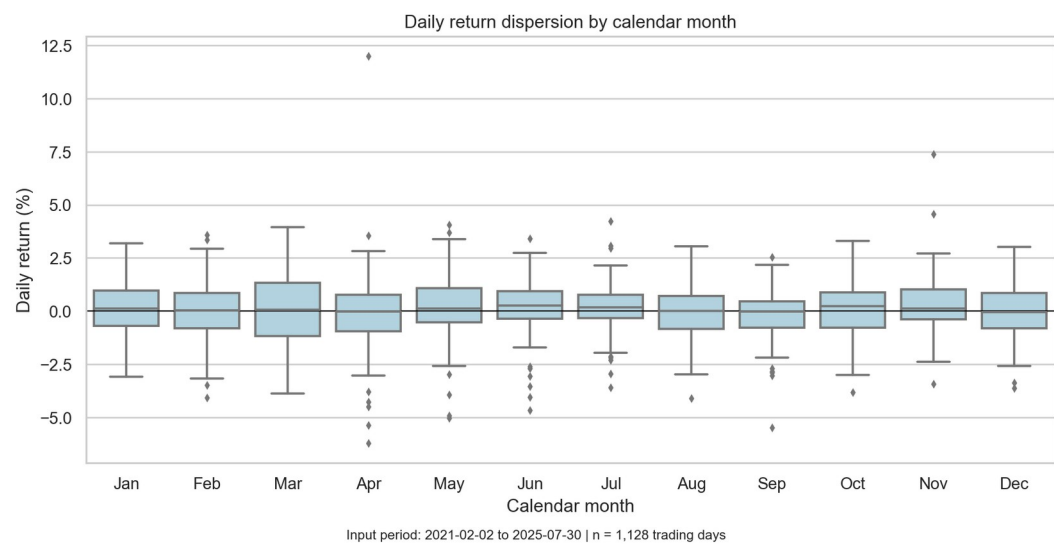


2

+12.0031%

0.0609%1.4621%—6.2109%

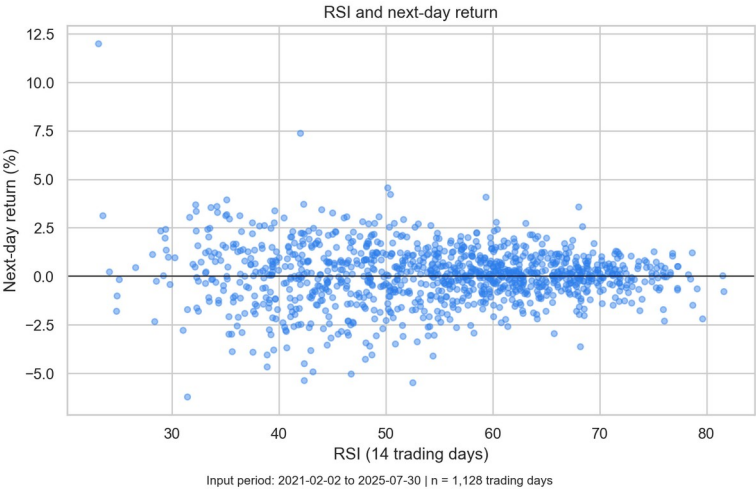
4



3 Box plot

Box plot

outliers



4 Scatter plot

RSI 14

RSI

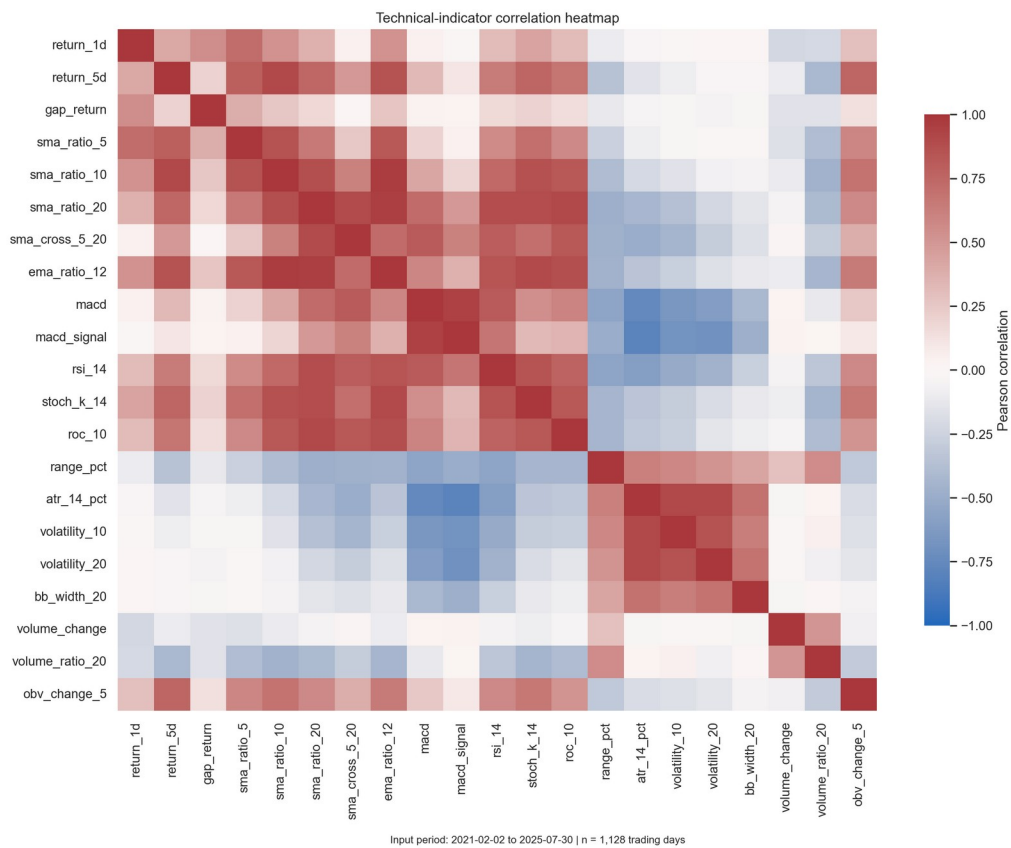
Pearson

RSI

next_return

-0.0312

4



5 Heatmap Pearson technical indicators

$r = 0.9506$ $r = 0.9694$ permutation importance [3]



6 Bar chart target training period

Up 614 (54.43%) Down or flat 514 (45.57%) balanced accuracy accuracy

5

Feature engineering 21 OHLCV t
return history 3, trend 7, momentum 3, volatility 5 volume 3 return_1d, SMA/EMA
ratios, RSI 14, ATR 14 volume ratio 20

Regression median imputation scaling Majority baseline , Logistic
min_samples_leaf 12, pipeline, Random Forest 700 max_depth 5
HistGradientBoosting 200 iterations learning_rate 0.03 output GradientBoosting

Train	1,128	2021-02-02 2025-07-30
Boundary gap	1	2025-07-31
Test	283	2025-08-01 2026-09-16

train/test label
test train expanding TimeSeriesSplit 5 folds gap 1 [2] random
shuffle
accuracy, balanced accuracy, precision, recall, F1 ROC AUC test Permutation importance
GradientBoosting feature 30 test balanced accuracy
[3]

6

	Accuracy	Balanced	F1	ROC AUC
Majority baseline	54.77%	50.00%	70.78%	0.5000
Logistic regression	50.88%	50.06%	56.70%	0.4806
Random forest	51.59%	50.98%	56.51%	0.5034
GradientBoosting	53.00%	52.74%	56.39%	0.5269

RQ3: Random Forest

accuracy 51.59%

baseline 54.77%

3.18

balanced accuracy

Random Forest

50.98%

ROC AUC 0.5034

baseline

GradientBoosting

balanced accuracy

52.74%

accuracy 53.00%

baseline

1.77

ROC AUC 0.5269

Cross-validation

GradientBoosting

train

balanced accuracy

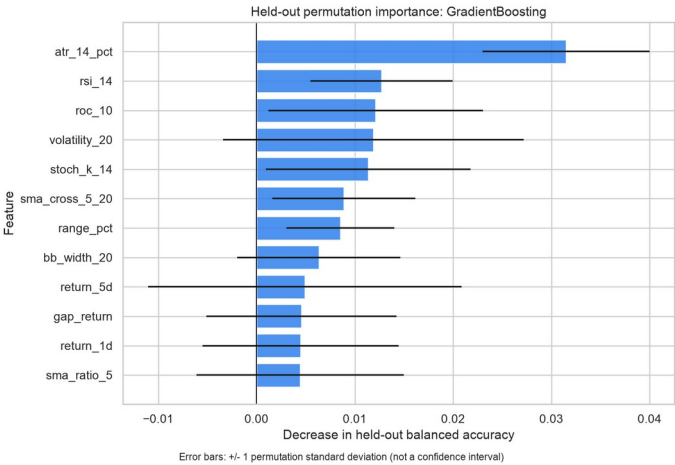
50.12% ± 2.37%

folds

ROC AUC

0.4965 ± 0.0115

holdout



7 Permutation importance

test

GradientBoosting

1 SD

RQ2: atr_14_pct

importance mean 0.0315 ± 0.0085

rsi_14 0.0127 ± 0.0072

roc_10 0.0121 ± 0.0109

volatility

importance 0.0577

importance

7

RQ1:	training					outliers
			49.89		EDA	
	RSI					
RQ2:	GradientBoosting	ATR 14		volatility	predictive association	
	permutation importance	test			metric	
RQ3: Random Forest		majority baseline	accuracy	283	GradientBoosting	balanced
accuracy	50%		CV	50%		
				walk-forward evaluation		
holdout		feature importance				
		adjusted/				
	OHLCV			Close		
	test				exploratory	

8

[1] Nasdaq. Invesco QQQ Trust Series 1 Historical Data. <https://www.nasdaq.com/market-activity/etf/qqq/historical> ; API request: <https://api.nasdaq.com/api/quote/QQQ/historical> (26 2026)

[2] scikit-learn developers. TimeSeriesSplit documentation. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html (26 2026)

[3] scikit-learn developers. Permutation feature importance documentation. https://scikit-learn.org/stable/modules/permutation_importance.html (26 2026)

[4] Big Data Analytics Project Assignment. [docs/Big_Data_Analytics_Project_Assignment.pdf](#)

: notebooks/pipeline_qqq_5_year.ipynb outputs/tables/5_year/; EDA
outputs/figures/5_year/